

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2021

Wednesday 26 May

Opening time 09:30 (BST)

You have 3 hours writing time to complete the paper and up to 30mins technical time to upload your answer files.

Mode of completion: Mixed mode. Please start the answer to each question in the relevant response box. Questions with sub-sections (e.g., Q1a & Q1b) should all be answered in the response box for the corresponding overall questions number.

There are 10 questions in this paper.

*Answer **all** questions from Part A and **two** questions from Part B.*

Part A attracts 40% of the marks in total; each question in Part B attracts 30% of the marks.

*In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;
in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.*

The limit for essays is 1600 words, with technical materials (sensibly sized diagrams and/or equations, etc.) contributing to the total at the rate of 350 words per A4 page.

Candidates may use their own calculators.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy with two goods, x and y , and two consumers, our friends Alice and Bob. Alice's utility function is $u_A(x_A, y_A) = x_A y_A^3$, while Bob's is $u_B(x_B, y_B) = 16 x_B y_B$. The initial endowment of Alice in this economy is $\omega_A = (8, 8)$, while the initial endowment of Bob is $\omega_B = (0, 0)$. Assume that good y is the numeraire and set $p_y = 1$.
 - (a) Discuss whether the allocation composed by consumptions $(x_A, y_A) = (2, 4)$ and $(x_B, y_B) = (6, 4)$ is the best according to the utilitarian criterion. Is it Pareto optimal?
 - (b) The social planner wants consumptions $(x_A, y_A) = (2, 4)$ and $(x_B, y_B) = (6, 4)$ to result from a competitive equilibrium of the economy, after some appropriate redistribution of the endowment. If the social planner desires to not use good x in this endeavour, how much of the numeraire good y has to be transferred from Alice to Bob?
 - (c) Alternatively, the social planner is considering to implement the Rawlsian solution. Give formal arguments to justify why, in this economy, the Rawlsian solution must necessarily be an allocation satisfying the following properties: (i) it equates the utilities of Alice and Bob, and (ii) it equates the marginal rates of substitution of Alice and Bob. Are there other necessary conditions?

[25%]

2. One of the most beautiful gardens in Oxford is maintained by College [A]. The college tries to improve what Mother Nature created by growing an amount y of yew trees. The utility of College [A] is $u_A(y, m_A) = 42y - y^2 + m_A$, where m_A denotes money of college [A] in pounds. Due to an old right of passage, members of College [B] can freely enjoy these beautiful gardens, and the utility of College [B] is $u_B(y, m_B) = \frac{1}{4}y^2 + 4y + m_B$, with m_B denoting money of college [B] in pounds.
- (a) Suppose that the production function for yew trees is $y = 2h$, where h is the number of gardening hours worked. If the cost of a gardening hour is 20 pounds, what is the amount of hours contracted by College [A], and the number of yew trees grown? What would be the Pareto efficient amount of yew trees grown? Assume that the initial monetary endowments of the colleges, before any yew trees have been planted, are m_A^* and m_B^* . If college [A] pays for all the yew trees planted in the two scenarios presented, compute the utilities of the two colleges under these two different scenarios and discuss whether either of them is Pareto superior to the other.
- (b) Compute all possible monetary transactions between colleges [A] and [B] that, being acceptable for both colleges, allow to implement the Pareto efficient amount of trees.
- (c) Discuss the structure of the utility functions of both colleges, and the role of this structure in your analysis.

[25%]

3. The inverse demand in a market is given by the function:

$$p(q) = \begin{cases} \alpha - q & \text{if } q \leq \alpha \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha > 0$ is a constant that reflects the attractiveness of the good that is sold in the market, p is the price of the good, and q is the total quantity of the good in the market. There are potentially two firms operating in this market, that compete in quantities. They have different marginal costs satisfying $\alpha > c_1 > c_2 > 0$ (that is, if the less-efficient firm 1 produces q_1 , the cost is $c_1 q_1$, and analogously for the more-efficient firm 2).

- (a) Compute the best response functions of both firms, by noticing that there could be two scenarios of relevance; namely, it can be that $\alpha - c_1$ is greater, or smaller, than $\frac{\alpha - c_2}{2}$. Draw the diagram of best responses in each of these two scenarios.
- (b) Under these two scenarios, discuss which the Cournot-Nash equilibrium of the game is, and which firm produces more output.

[25%]

4. A worker has a utility function of the form $u(w, e) = \sqrt{w} - e$, where w describes monetary wage and e is a measure of the effort put in working activities. Currently, the best job available for this worker involves a simple contract that requires from him to do colossal effort (measured as $e = 2$) for a fixed wage of 36 units.

A risk-neutral principal is initiating a project and wants to hire this worker. This project can bring two possible gross profit outcomes to the principal, $\pi_L = 30$ and $\pi_H = 70$, but the probability of these outcomes depends on the effort level put by the worker. Given the characteristics of the project, there are two relevant levels of effort that can be put by the worker: basic effort (measured as $e = 0$) or substantial effort (measured as $e = 1$). If the worker exerts basic effort, the probability of π_H is only $1/4$, but if the worker exerts substantial effort, the probability of π_H increases to $1/2$.

- (a) What is the reservation utility of the worker? Now, assume that effort is observable in the project that will be initiated. What wage will the principal offer if she wants to hire him and induce basic effort? What wage will she offer if she wants to hire him and induce substantial effort? What contract is optimal for the principal?
- (b) Now assume that effort is unobservable in the project that will be initiated. The principal considers contracts of the form $(w(\pi_L), w(\pi_H))$ where $w(\pi_j)$ is the wage paid by the principal to the agent if the gross profit is π_j . What is the optimal way for the principal to induce low effort? What is the optimal way for the principal to induce high effort? What is the expected wage and the agency cost in this case? What is the optimal contract?

[25%]

Part B

5. A small rural community wants to select one out of a number of proposals. Start by suggesting an intuitive real-life situation on which all the community members may reasonably have single-peaked preferences over the proposals at stake. For the example that you introduce, discuss what relevant factors could be considered by the community while deciding which mechanism to use in order to reach an optimal choice. Analyse formally one reasonable mechanism that could be used, with its main advantages and disadvantages.
6. The UK government is thinking about providing for free a new Covid-19 app dedicated to improving the mental health and well-being of UK residents. Would such an app qualify as a public good? Why or why not? What funding mechanisms (e.g., use of taxation, crowdfunding, etc.) should the government use to ensure that the cost of the app is covered and why?
7. Is it possible to justify the use of the Nash equilibrium concept as a predictor of how a game will be played by simply appealing to players' rationality? If not, how else can you justify the usefulness of Nash Equilibrium? Given this, do you think that Nash Equilibrium is a reasonable solution concept?
8. Explain how expected utility is constructed using a utility function u over money. What property of u corresponds to the decision maker being *risk averse*? What does it mean for one decision maker to be *more risk averse* than another and how would that be reflected in their respective utility functions? What does it mean, in terms of behaviour, that a person exhibits *decreasing absolute risk aversion* (DARA) and what properties would the utility function of such a person exhibit? Discuss whether the observation of elite professionals in sport insuring themselves against injury is a reason to reject the DARA hypothesis.
9. Consider an educational signalling model. In two otherwise identical countries, you observe different equilibria. Country A is in a pooling equilibrium, while country B is in a separating equilibrium. Can you say which country faces greater welfare losses? Can we say which type of equilibrium is preferred by the least able agents in these countries? And by the most able agents?

10. Goku is trying to decide which crop to grow in his field: wheat, rice or soybeans. The following table summarises the monetary returns when each of these crops is grown, depending on the weather conditions within the region. Goku knows the probability of each possible weather condition, information that is also described in the table.

Weather	Good	OK	Bad
Probability	.4	.3	.3
Return, Wheat	36	36	25
Return, Rice	64	36	9
Return, Soybeans	100	36	0

- (a) Write down the monetary lottery generated by each crop, assuming that returns in the table are just total wealth. Show that soybeans have the largest expected return. If Goku is an expected utility maximiser with monetary utility $u(x) = \sqrt{x}$, show that he would choose to grow rice. Explain this divergence. [25%]
- (b) If the OK-returns of all crops were 30 instead of 36, would Goku still grow rice? Use the property of independence or the additive nature of expected utility in your argument. [15%]
- (c) Goku confirms that OK-returns are 36 and decides to visit Cro-op. This insurance company offers a plan to help growing soybeans. Contracting with Cro-op has an unconditional cost of 21 units for the farmer and, only if weather conditions are bad, the company makes a payment of 36 units. Write down the lottery for this plan and show that Goku will not contract the plan. [10%]
- (d) Goku contacts Ropporu, who is like Goku in all respects and has a field in the same region. Notice that, being in the same region, weather is perfectly correlated for both fields and both farmers will suffer, or enjoy, the same realised weather. Explain why the farmers can improve their situation by growing soybeans in one of the fields and wheat in the other, sharing equally the returns. [15%]
- (e) Given the good prospects of growing a combination of wheat and soybeans and sharing returns, a group of farmers in the region, all of them equal to Goku, decide to cooperate. Denoting by λ the fraction of fields in which wheat is planted (hence, $1 - \lambda$ will be the fraction of fields in which soybeans are planted), write down the maximisation problem that the group faces, derive its first-order condition, and identify the optimal value of λ . [20%]
- (f) Finally, the government makes it mandatory to grow wheat. Discuss briefly why sharing profits with Ropporu or other farmers in the region cannot improve their situation. However, Mulan is also equal to Goku in all respects (including the weather probabilities) but lives in a very distant land. Discuss using theoretical concepts whether, and why, an agreement to share profits would help both Goku and Mulan now that risks are independent. [15%]